Fabrication and Testing of AC-coupled Low-Gain Avalanche Diodes at BNL

Gabriele Giacomini, Wei Chen, Gabriele D'Amen, Luigi Lavitola, Alessandro Tricoli

***Abstract*– We present the fabrication and the first functional tests on a new class of silicon devices: the AC-coupled Low-Gain Avalanche Diodes (AC-LGAD). Because of its good timing performance, the LGAD was originally developed to support the silicon tracker at High-Energy Physics experiments for the reconstruction of pile-up events originating from a same bunch crossing. Pile-up events happen in fact at slightly different times and adding a fourth dimension (the time) is crucial to resolve them. On the other hand, LGADs suffers from poor spatial resolution. AC-LGADs offer a solution to this drawback, while retaining the good timing performance. Test structures have been fabricated at BNL, using the internal silicon processing facility, and have been tested with fast electronics and radioactive sources. Large gains and fast signal have been measured, while the noise stays low and comparable to the standard LGAD case. Other wafer fabrications are ongoing to further improving performances and correct pitfalls in the very first fabrication.**

I. Introduction

Low-Gain Avalanche Diodes (LGADs) are a class of silicon devices which have recently attracted attention in the High Energy Physics (HEP) community for their fast-timing performance [1]. Fabricated on a thin high- or medium resistivity *p*-type substrate, they feature a low-doping *p*-implant (usually called "*gain layer*") below a *n*+ implant, the latter making the junction with the substrate (Figure 1a) [2]. Application of a bias voltage to the device leads to depletion of the *p* gain layer, creating for a depth of about one micron an electric field intense enough to trigger impact ionization for the electrons crossing this superficial region. The impact ionization of the holes is instead negligible (if the electric field is low enough), thus the onset of an avalanche breakdown is excluded. The signal is dominated by the motion of the multiplied holes in the thin substrate. This boosted signal develops in about one nanoseconds and timing resolution of a few tens of picoseconds has been demonstrated, due to the large signals and the thin substrate. Typical gains are in the range of 10-50. A condition to have a uniform multiplication across the full area of the device is that such area must be much larger than the thickness of the substrate. For example, in the High Granularity Timing Detector (HGTD) of ATLAS at the High Luminosity Large Hadron Collider (HL-LHC), LGAD pads are about 1.3 mm x 1.3 mm. So, if on one hand the timing is very good, on the other hand the spatial resolution is poor. A way to circumvent this problem is to place the *p*+ pixels or strips, with the desired granularity, on the opposite side of the wafer. Being the substrate very thin to preserve fast timing (order of 50 µm), and due to impossibility to process in a clean room silicon wafers thinner than 200µm (for 4in wafers), it is necessary to remove (in part or totally) the substrate by grinding, thinning or etching and then pattern the electrodes. This double-sided approach, if feasible, is very complicated. Another approach is to exploit the AC-coupled LGAD option [3]. In this structure, the large *n*+ at the interface is replaced by an equally large *n*- resistive layer (Figure 1b), much less doped and thus much more resistive. Still this layer can provide depletion of the gain layer (and thus creating the high electric field for impact ionization), while keeping electric field lines parallel and therefore assuring a uniform multiplication. Metal pads are placed over a dielectric insulator and are insulated from the silicon underneath. Carriers drifting in the substrate induce signal on these floating metal pads, AC-coupled with the *n*-resistive layer.

Brookhaven National Laboratory (BNL) is active in the development of standard LGADs [4] and is now investigating this new approach [5].

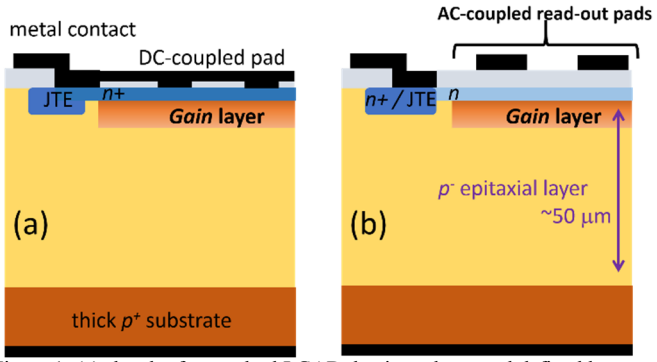


Figure 1. (a) sketch of a standard LGAD, having a large pad defined by an *n*+ over a gain p^- layer, and contacted by a metal pad; (b) structure of an AC-LGAD, where the *n*+ is replaced by an *n*-resistive layer, and the metal pads are patterned over a dielectric.

Manuscript received December 13th, 2019. This work was supported in part by the U.S. Department of Energy under Grant No. DE-SC0012704. This research used resources of the Center for Functional Nanomaterials and of the National Synchrotron Light Source II, which are U.S. Department of Energy (DOE) Office of Science User Facilities operated for the DOE Office of Science by Brookhaven National Laboratory under Contract No. DE-SC0012704.

G. Giacomini, W. Chen, G. D'Amen, and A. Tricoli are with Brookhaven National Laboratory, Upton, 11973, NY, USA, e-mail: giacomini @bnl.gov. L. Lavitola is with Universita degli Studi Federico II, Naples, Italy.

978-1-7281-4164-0/19/$31.00 ©2019 IEEE

Authorized licensed use limited to: Institute of High Energy Physics. Downloaded on December 04,2024 at 01:53:42 UTC from IEEE Xplore. Restrictions apply.

II. FABRICATION

For this, we modified the LGAD process by replacing the *n*+ with a much less doped *n* layer, while an *n*+ frame is still implanted at the border of the active area to provide the contact through which the electron current leaves the device. A high-resistance *n* is needed so that most of the signal is fed to the read-out electronics through the coupling capacitance C. The latter is made by the metal pad separated from the *n* by a dielectric (a thin layer of silicon nitride, in our case). The condition to have most of the signal fed to the read-out electronics is that, at the read-out frequency ω, RC >> ω, where R describes the resistance of the *n* layer. It is important, thus, to maximize both the coupling capacitance (by using a thin insulator) and the resistance of the *n* layer (by decreasing the implantation dose of the *n* layer). Test structures with metal pads of different dimensions (and thus different capacitance towards the *n*-resistive layer) have been inserted in the new design (for example, Figure 2), for which new photolithographic masks for the metal and the contacts have been used.

Due to the not-negligible depletion of the low-dose *n* resistive layer, the gain layer is expected to result deeper (as measured in a C-V) and this will affect the impact ionization integral. A lower implantation dose for the p+ gain layer is thus needed to prevent an early breakdown.

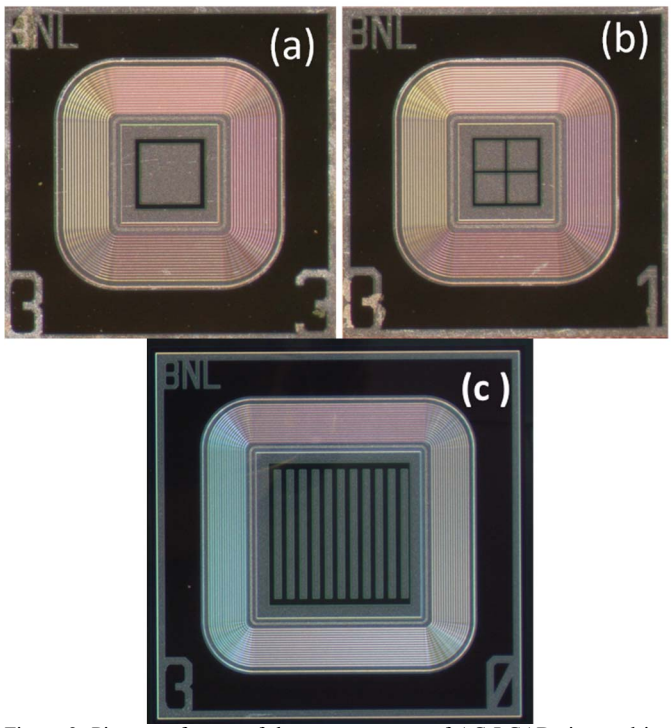


Figure 2. Pictures of some of the test structures of AC-LGADs inserted in a dedicated wafer. (a) Single pad; (b) 2x2 pad array, (c) strip sensor.

III. FUNCTIONAL CHARACTERIZATION

After fabrication, a few samples were diced out from the wafer and mounted on a multichannel printed circuit boards (PCB), made by Fermilab (Figure 3), populated with 16 independent low-noise fast transimpedance amplifiers (TA). Devices include small arrays of pixel and strip sensors, as shown in Figure 4. Before connecting the AC-coupled pads of these devices to independent channels of the TA board, we connected the (DC-coupled) *n*+ at the border of the device to a TA channel. By doing this, we could test the performance of the DC-coupled LGAD made by the same doping profiles and biased at the operational voltage. We can thus compare the performance of the AC-LGAD and of the LGAD having exactly the same gain. Signals generated by the interaction of βs from a ^{90}Sr source were acquired by a 1GHz scope. An example of such a waveform is shown in Figure 5, for an AC-LGAD pixel (pixel pitch 250 µm). Waveforms are very similar to those from a regular DC-coupled LGAD, they are very fast, and exhibit an undershoot (of the order of tens of nanoseconds). The total integral of the signal is zero, due to the fact that the total charge collected by the AC pad is null. By acquiring pulse height spectra from an AC-LGAD channel and the LGAD, we saw that their distributions were very similar, peaking at the same channel. Most of the signal is thus induced on the read-out electronics, meaning that for our pixel geometry the condition of RC>> ω holds [5].

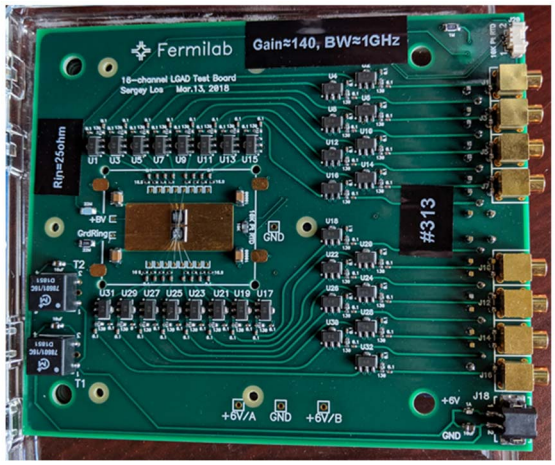


Figure 3. Multichannel TA board made by Fermilab, with AC-LGAD test structure connected to its inputs.

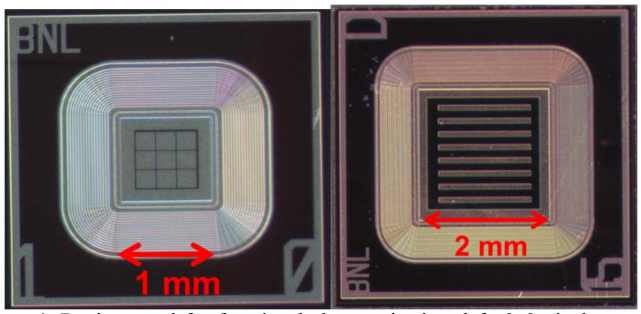


Figure 4. Devices used for functional characterization: left, 3x3 pixel array, with pixels at 250µm pitch; right, strip sensor made of 8 1.5 mm long strips at a pitch of 200 µm.

Authorized licensed use limited to: Institute of High Energy Physics. Downloaded on December 04,2024 at 01:53:42 UTC from IEEE Xplore. Restrictions apply.

We applied the Transient Current Technique method to map the sensor response to an IR or red laser, impinging perpendicularly to the surface of the device. The laser power is set in order to generate the same charge produced, in average, by a mip in silicon. Multiple signals generated in the same spot are averaged, to minimize the contribution of the noise. As an example, a result of such a scan is shown in Figure 6, where the response of the second strip from the top as a function of the hitting position is shown. This method is particularly useful to measure the crosstalk between strips. If the laser shines in between two strips, these two strips obviously share the charge; a nearby strip sees 13% of the signal seen by either one of the two strips close to the event, this percentage then drops to 6% and 4% if we read-out strips more and more distant from the hit position. These numbers are very dependent on the parameters of the sensor (value of the coupling capacitance and the resistance of the *n* layer, as well as geometrical) and suggest that a fine tuning of these parameters can be used to decrease or increase the cross-talk (the latter could be useful, for example, for interpolation through peak or charge measurements).

Notice how, if the laser shines close to the border where the *n*+ implant is present, an important percentage of the signal is cross-coupled back to very distant strips. This can be explained by the fact that this n+ layer acts as a low resistance path, which helps the cross-coupling of signal to distant electrodes.

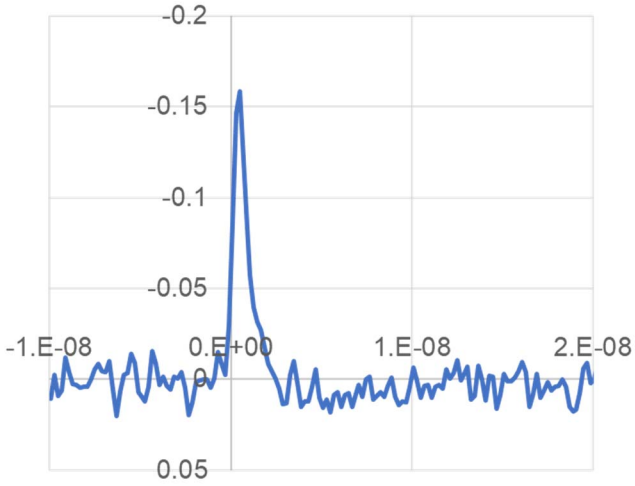


Figure 5. Waveform as acquired from the 1GHz scope, generated by betas from ^{90}Sr impinging the 9-pad AC-LGAD of Figure 4.

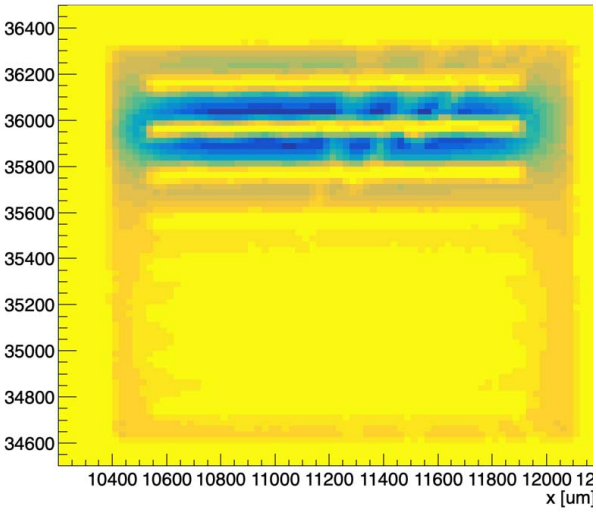


Figure 6. Map generated by scanning an IR laser perpendicular to the surface of an AC-LGAD, whose metal AC-pads are patterned in strips (Figure 4), and measuring the pulse height from one of the strip.

IV. CONCLUSIONS

Based on our expertise on the fabrication of LGADs, we fabricated prototypes of AC-coupled LGADs, by changing a few photolithographic masks and modifying some process steps. The advantage of the AC-LGAD approach is that fine-pitch electrodes, excluded in the LGAD case, are now possible. Functional measurements performed with radioactive sources show signals that are as fast and have the same amplitude as signals from LGADs.

REFERENCES

[1] H. F-W Sadrozinski et al., "4D tracking with ultra-fast silicon detectors," 2018 Rep. Prog. Phys. 81 026101, (doi: https://doi.org/10.1088/1361-6633/aa94d3).

[2] G. Pellegrini et al., "Technology developments and first measurements of LowGain Avalanche Detectors (LGAD) for high energy physics applications," Nuclear Instruments and Methods in Physics Research A765 (2014) 12–16, http://dx.doi.org/10.1016/j.nima.2014.06.008.

[3] M. Mandurino, "Resistive AC-Coupling: a new readout paradigm in 4D tracking with Ultra-Fast Silicon Detectors," 13th "Trento" Workshop – Munich, 19-21 February 2018,

[4] G. Giacomini et al., "Development of a technology for the fabrication of Low-Gain Avalanche Detectors at BNL," arXiv:1811.04152 [physics.ins-det].

[5] G. Giacomini et al.: "Fabrication and performance of AC-coupled LGADs," 2019 JINST 14 P09004.

Authorized licensed use limited to: Institute of High Energy Physics. Downloaded on December 04,2024 at 01:53:42 UTC from IEEE Xplore. Restrictions apply.